

Joshua Koshy

CS 499 Milestone Two Narrative

Artifact Description

The artifact I selected for enhancement is the Security Threat Monitoring Dashboard. The original version of the artifact was made of a backend application developed using Node.js, Express, MongoDB Atlas, and Mongoose. The system supported user authentication through JSON Web Tokens (JWT), allowed security logs to be created and stored, and automatically generated alerts based on security events. The original artifact functioned primarily as a backend API and was tested through Postman.

Enhancement Overview

For the software design and engineering enhancement, I expanded the artifact into a more complete full stack application. The biggest enhancement was the addition of a frontend dashboard that allows users to interact with the system through a graphical interface instead of relying solely on API testing tools. The dashboard provides a login interface, displays security metrics, allows users to create security logs, and presents generated alerts in an organized format.

Other enhancements included improved input validation and error handling. Validation was added to make sure that required fields are provided during user registration, login, and security log creation. These improvements help prevent invalid data from entering the system and improve the overall reliability of the application. Documentation was also improved through additional comments and an updated README file to support maintainability and future development.

Software Design and Engineering Concepts Demonstrated

This enhancement shows several software engineering principles. The application uses a modular architecture that separates routes, middleware, database models, configuration files, and frontend components. This organization improves maintainability and scalability. Security best practices were added through password hashing with bcrypt, token-based authentication using JWT, and server side validation.

The addition of a frontend dashboard also shows a full-stack integration by connecting the user interface to backend API endpoints. The dashboard retrieves and displays data from the backend while providing users with a more intuitive way to interact with the system.

Reflection

Completing this enhancement improved my understanding of software design and engineering concepts related to full stack application development. I gained experience connecting frontend and backend components as well as implementing validation and error handling, and improving the overall usability of an application. These enhancements transformed the artifact from a

backend only API that had some logic into a more complete security monitoring system that better reflects professional software engineering practices. The project also provides a strong foundation for future enhancements involving algorithms, data structures, databases, cloud deployment, and advanced threat analysis.